An Evolutionary AI Framework for Forest Cover Monitoring and Vegetation Analysis Using Sentinel-2 Satellite Data

Keywords - Forest cover monitoring, vegetation, wildlife safety, Wayanad, Land slide, anomaly, Monsoon based natural disaster, self-supervised learning, Genetic algorithm, sentinel-2 data

Contents

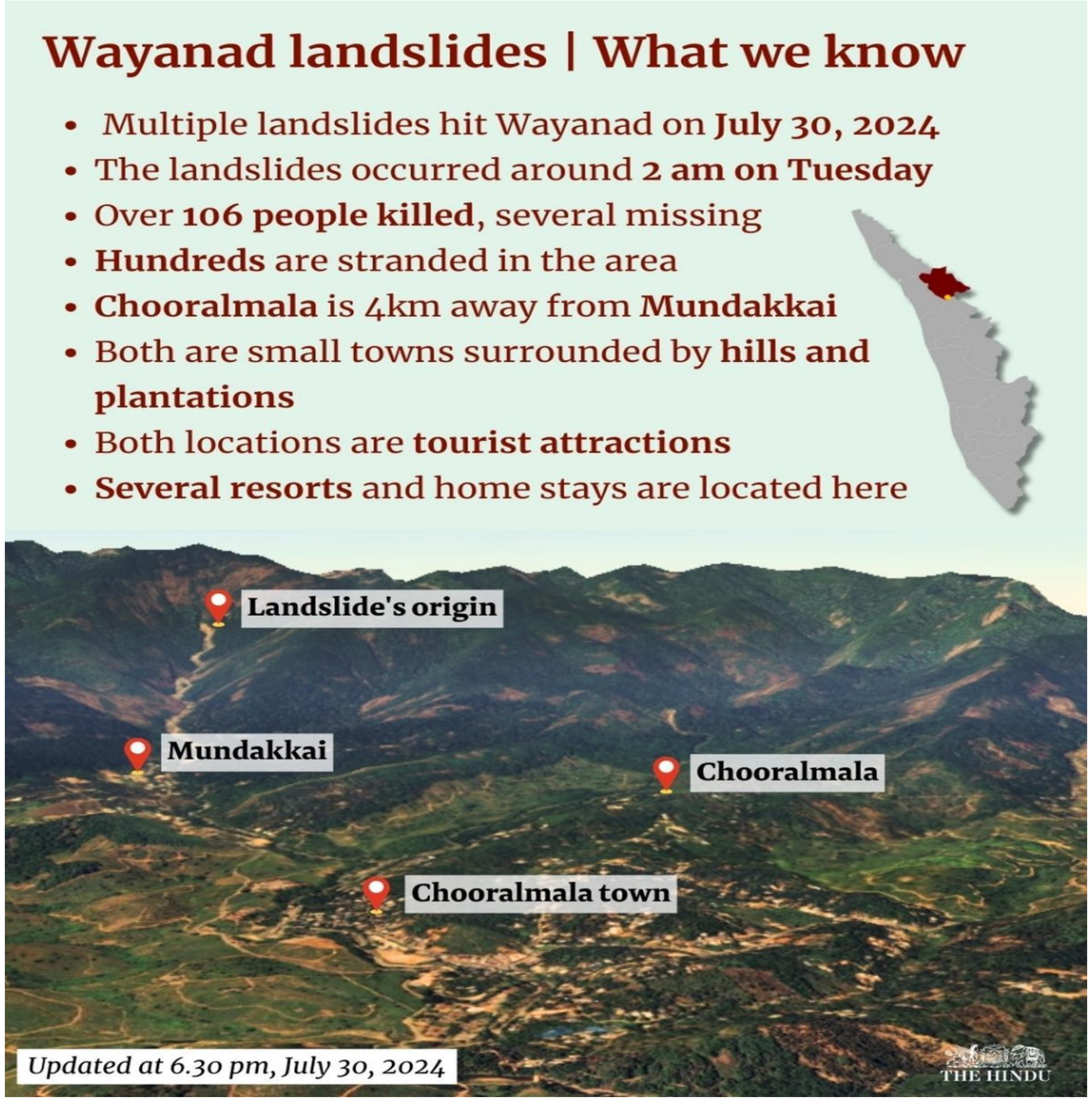
- Introduction, motivation, literature study, research gap, problem statement & objectives, methodology, plan of action for two semesters..

Questions to address:

- Complexity of the algorithm
- Compute requirements – check with T4 (Colab)
- Land location dependency
- Existing methods for landslide prediction + integrate with monsoon forecast + data or metric pattern from the sentinel 2
- Computational time -?

Introduction

- Forest degradation begins long before deforestation becomes visible. Gradual changes caused by illegal logging, disease, climate change, and extreme monsoon events reduce forest health, but these early warning signs often remain unnoticed until significant damage has already occurred.
- Existing monitoring methods rely on field surveys and periodic reports such as the India State of Forest Report (ISFR), which provide limited temporal coverage. Although Sentinel-2 satellite imagery enables continuous large-scale observation, most AI-based approaches require extensive labeled datasets and depend on fixed thresholds that fail to adapt to seasonal and regional variations.
- To address these challenges, EvOLve introduces an Evolutionary AI framework that combines self-supervised learning with Genetic Algorithm optimization to learn from largely unlabeled satellite imagery, continuously adapt its detection thresholds, and provide early, reliable identification of forest degradation for timely conservation and disaster prevention.



Motivation

Literature survey

Research gap

Problem statement & Objectives

Methodology

Dataset

- Sentinel-2
- Data from Jan-2019 till Dec-2025.

Plan of action – sem 7

Plan of action – sem 8